



दक्षिण बिहार केन्द्रीय विश्वविद्यालय CENTRAL UNIVERSITY OF SOUTH BIHAR

Question Paper of End-Term Examination May 2025

Name of Department	:	Computer Science	Programme	:	M.Sc. (AI)
Session	:	2024-2025	Semester	:	II
Course Title	:	R Programming	Course Code	:	CAI82ME03900
Course Credit	:	Non-Credit	Max. Marks	:	70
			Duration	:	2 hours
Enrolment No. (to be filled by Student)	:CUSB2402332028				

Instructions: This question paper have three sections. Attempt each section.

Section A

This section has five questions of two points each. Attempt all questions.

Q1. Write the syntax in R programming language to compute the following.

$$(4 \times 6)^2 + 5$$

Q2. Name any one R-package along with one of its function.

Q3. Write a command in R to find out data type of a variable.

Q4. Write the command in R to remove the existing variable in R.

Q5. Write the command to load the data frame "economics".

Section B

This section has five questions of ten points each. Attempt any four questions.

Q1. Explain the following data type in R along with example.

I. Logical data type

II. Character data type

- Q2. Explain the Vector in R. Write the command in R to print the square root of 20 natural numbers in R using Vector.
- Q3. What are the NA and NULL in the missing data handling in R. What is the utility of the R function *is.na(z)*; where z is a vector.
- Q4. Write the syntax of while loop in R with the help of an example.
- Q5. Write a short note on usability of R programming language.

Section C

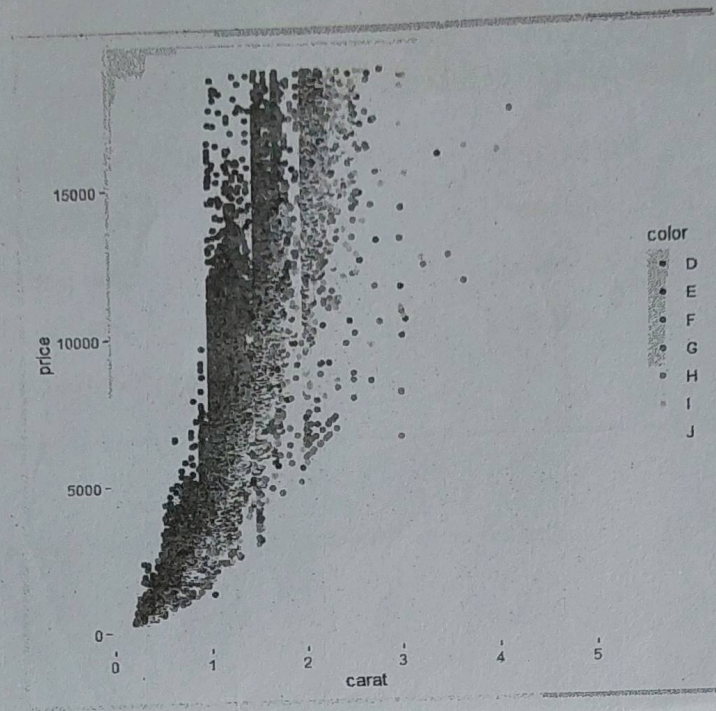
This section has two questions of twenty points each. Attempt any one question.

- Q1. What is Data Frame in R. Explain the following functions in terms of the advanced data structure Data Frames.

[2+9×2]

- I. `nrow()`
- II. `dim()`
- III. `head()`
- IV. `names()`
- V. `class()`
- VI. `ncol()`
- VII. `tail()`
- VIII. `rownames()`
- IX. `data.frame()`

Q2. Write the R codes to print following scatter plot on the diamonds table (consisting of 53000 rows) available under the ggplot2 package. Assume that ggplot2 is not installed at initial stage. Also explain purpose of each command used in comment section.



This question is solved by
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